

# Sample Technical Documentation Pack

User Manual Excerpts, Maintenance & Parts, Troubleshooting Guides, FAQs and Resolution Procedures

SYNTHETIC TEST DATA - NOT REAL MANUFACTURER DOCUMENTATION

This document is entirely fictional. It was written to test an AI customer-support assistant (speech, sentiment, categorisation and RAG suggestions) before real client manuals are available. Machine models, part numbers, error codes, settings and procedures are invented and must never be used to operate, maintain or repair real equipment. It contains no client or manufacturer data.

| Machine | Industry      | Fictional model               |
|---------|---------------|-------------------------------|
| WP-400  | Woodworking   | CNC Panel Saw                 |
| SC-900  | Stone cutting | Bridge Saw                    |
| MC-250  | Metal cutting | CNC Plasma Cutting Table      |
| GC-120  | Glass cutting | Automatic Glass Cutting Table |

Document map

| Part | Content                                                     | Support category                                |
|------|-------------------------------------------------------------|-------------------------------------------------|
| 1    | User manual excerpts (startup, settings, normal operation)  | Machine Operation Issues                        |
| 2    | Maintenance schedules, parts catalog, ordering and warranty | Maintenance & Parts                             |
| 3    | Error codes and symptom tables                              | Technical Troubleshooting                       |
| 4    | Step-by-step resolution procedures                          | Technical Troubleshooting / Maintenance & Parts |
| 5    | Frequently asked questions                                  | Mixed                                           |

# PART 1 - USER MANUAL EXCERPTS

## 1.1 WP-400 CNC Panel Saw (Woodworking)

### Section 1.1.1 Daily startup procedure

*Doc ref: UM-WP400-1.1.1 | Category: Machine Operation Issues | Machine: WP-400*

1. Check the compressed-air gauge. Working pressure must read 6.0 bar (0.6 MPa). Do not start below 5.5 bar.
2. Inspect the main saw blade for missing teeth, cracks or resin build-up. Replace a damaged blade before use.
3. Turn the main isolator to ON. The controller boots in about 45 seconds.
4. Press HOME. All axes (X, Y, blade height) must complete homing without alarm.
5. Start the dust extraction. The blade cannot start while extraction is off (interlock).
6. Load a cutting program and run one test cut on scrap before production.

### Section 1.1.2 Selecting feed rate and spindle speed

*Doc ref: UM-WP400-1.1.2 | Category: Machine Operation Issues | Machine: WP-400*

Use the table as a starting point. If the cut edge chips, reduce the feed rate by 10-20% before changing anything else.

| Material           | Thickness | Spindle speed | Feed rate |
|--------------------|-----------|---------------|-----------|
| Melamine chipboard | 18 mm     | 4000 rpm      | 20 m/min  |
| MDF                | 18 mm     | 4000 rpm      | 18 m/min  |
| Plywood            | 12 mm     | 4000 rpm      | 22 m/min  |
| Solid hardwood     | 30 mm     | 3500 rpm      | 8 m/min   |

### Section 1.1.3 Dust extraction

*Doc ref: UM-WP400-1.1.3 | Category: Machine Operation Issues | Machine: WP-400*

Minimum air velocity at the saw hood is 20 m/s. The controller raises alarm E-102 when the filter pressure drop exceeds 1200 Pa. Clean or replace the filter cartridge WP4-FL-DE1 when the alarm appears. Never bypass the extraction interlock.

## 1.2 SC-900 Bridge Saw (Stone)

### Section 1.2.1 Startup and water supply

*Doc ref: UM-SC900-1.2.1 | Category: Machine Operation Issues | Machine: SC-900*

1. Open the water supply valve. Minimum flow is 8 L/min at 2 bar. Never run the blade dry.
2. Confirm the flow indicator on the operator panel is green.
3. Clamp the slab and check that it is fully supported underneath.
4. Switch on the spindle and wait 10 seconds for full speed, then lower the bridge.

### Section 1.2.2 Making a straight cut in slab

*Doc ref: UM-SC900-1.2.2 | Category: Machine Operation Issues | Machine: SC-900*

Set the blade depth 5 mm below the slab thickness. Recommended cutting speeds: granite 30 mm at 1.2 m/min, marble 30 mm at 2.0 m/min. For slabs thicker than 40 mm make two passes of equal depth. Slow down when the blade exits the slab to avoid chipping the underside.

## 1.3 MC-250 CNC Plasma Cutting Table (Metal)

### Section 1.3.1 Startup and gas supply

*Doc ref: UM-MC250-1.3.1 | Category: Machine Operation Issues | Machine: MC-250*

The plasma supply needs clean, dry compressed air at 6.2 bar (90 psi). Drain the filter bowl daily. Attach the work (ground) clamp directly to clean bare metal on the plate; paint and rust prevent a stable arc. Switch on the power

source, wait for the READY lamp, then home the torch height controller (THC).

### Section 1.3.2 Cut settings

*Doc ref: UM-MC250-1.3.2 | Category: Machine Operation Issues | Machine: MC-250*

Pierce height is 3.8 mm and pierce delay 0.4 s for mild steel. The THC arc-voltage target is 125 V.

| Mild steel | Current | Cut speed   | Cut height |
|------------|---------|-------------|------------|
| 3 mm       | 45 A    | 4500 mm/min | 1.5 mm     |
| 6 mm       | 65 A    | 3000 mm/min | 1.5 mm     |
| 10 mm      | 85 A    | 1700 mm/min | 1.5 mm     |
| 16 mm      | 105 A   | 900 mm/min  | 1.5 mm     |

## 1.4 GC-120 Automatic Glass Cutting Table (Glass)

### Section 1.4.1 Loading and scoring

*Doc ref: UM-GC120-1.4.1 | Category: Machine Operation Issues | Machine: GC-120*

The GC-120 handles sheet glass 3-12 mm thick. Set the scoring pressure for the thickness: 4 mm = 2.2 bar, 6 mm = 3.0 bar, 10 mm = 4.2 bar. Keep the cutting-oil level above the MIN mark. Vacuum pressure must read at least -0.6 bar before scoring.

### Section 1.4.2 Break-out and edge quality

*Doc ref: UM-GC120-1.4.2 | Category: Machine Operation Issues | Machine: GC-120*

Break the sheet within 30 seconds of scoring; the score line loses effect as the glass relaxes. A clean score is thin and continuous. Shells or flares along the edge mean the wheel is worn or the pressure is too high.

# PART 2 - MAINTENANCE & PARTS

## Section 2.1 Maintenance schedules

Doc ref: MP-ALL-2.1 | Category: Maintenance & Parts | Machine: All

| Machine | Daily                                                                               | Weekly                                                                   | Monthly / by hours                                                                                                    |
|---------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| WP-400  | Clean dust from table and rails.<br>Check air pressure (6.0 bar).<br>Inspect blade. | Grease linear rails with grease LG-2 (WP4-GR-LG2): 2 pumps per carriage. | Check spindle belt tension and clean extraction filter. Replace spindle belt WP4-BT-85 every 500 hours.               |
| SC-900  | Flush slurry from the tray. Check blade flange for damage.                          | Clean water filter element SC9-WF-02.                                    | Grease bridge rails. Check blade flange runout (limit 0.1 mm). Replace spindle bearings SC9-BR-6207 every 1000 hours. |
| MC-250  | Drain air filter bowl. Inspect consumables.                                         | Clean rails and rack. Check ground clamp cable.                          | Inspect torch lead. Replace air filter element MC2-AF-10 every 6 months.                                              |
| GC-120  | Wipe table. Check oil level and vacuum pressure.                                    | Clean the scoring head.                                                  | Lubricate guide rails. Replace scoring wheel after about 5000 m of scoring or when edge quality drops.                |

## Section 2.2 Lubrication

Doc ref: MP-ALL-2.2 | Category: Maintenance & Parts | Machine: All

Use only the lubricants named here. Do not mix greases. WP-400 and SC-900 rails: grease LG-2. GC-120 guide rails: light machine oil. Never lubricate plasma torch parts with petroleum grease; use silicone grease on torch O-rings only.

## Section 2.3 Parts catalog

Doc ref: MP-ALL-2.3 | Category: Maintenance & Parts | Machine: All

| Part No.    | Description                              | Fits           | Replace when                                   |
|-------------|------------------------------------------|----------------|------------------------------------------------|
| WP4-BL-300  | Saw blade 300 mm, 72 teeth, carbide      | WP-400         | Edge chipping or about every 80-120 hours      |
| WP4-BL-120  | Scoring blade 120 mm                     | WP-400         | Score line ragged or after 150 hours           |
| WP4-BT-85   | Spindle drive belt                       | WP-400         | Every 500 hours or if cracked                  |
| WP4-FL-DE1  | Dust extraction filter cartridge         | WP-400         | Alarm E-102 or every 3 months                  |
| WP4-GR-LG2  | Linear rail grease LG-2 cartridge        | WP-400, SC-900 | Used weekly                                    |
| SC9-DB-350  | Diamond blade 350 mm, segmented, granite | SC-900         | Segments below 3 mm height                     |
| SC9-DB-400  | Diamond blade 400 mm, segmented, marble  | SC-900         | Segments below 3 mm height                     |
| SC9-WF-02   | Water filter element                     | SC-900         | Alarm F-01 or monthly                          |
| SC9-BR-6207 | Spindle bearing 6207 (pair)              | SC-900         | Every 1000 hours or noise                      |
| SC9-SL-90   | Water seal kit                           | SC-900         | Water leak at spindle                          |
| MC2-EL-65   | Electrode, 65 A                          | MC-250         | Pit depth reaches 1.5 mm                       |
| MC2-NZ-65   | Nozzle, 65 A                             | MC-250         | Orifice oval or damaged                        |
| MC2-SH-65   | Shield cap, 65 A                         | MC-250         | Spatter damage                                 |
| MC2-SW-65   | Swirl ring, 65 A                         | MC-250         | Replace with every electrode change if cracked |
| MC2-AF-10   | Air filter element                       | MC-250         | Every 6 months                                 |
| MC2-TH-HT   | Torch height sensor                      | MC-250         | Error P-22 persists                            |
| GC1-SW-6    | Scoring wheel, carbide, 6 mm series      | GC-120         | About 5000 m or poor edge                      |

| Part No.  | Description             | Fits   | Replace when                  |
|-----------|-------------------------|--------|-------------------------------|
| GC1-OL-1L | Cutting oil, 1 litre    | GC-120 | Oil below MIN mark            |
| GC1-BP-04 | Break-out bar pad       | GC-120 | Pad worn or cut               |
| GC1-VB-02 | Vacuum table seal strip | GC-120 | Alarm G-03 and no other cause |

## Section 2.4 Ordering parts and warranty

*Doc ref: MP-ALL-2.4 | Category: Maintenance & Parts | Machine: All*

Order parts through your authorised distributor and quote the machine model, serial number and part number from Section 2.3. Wear parts (blades, electrodes, nozzles, scoring wheels, filters, belts) are consumables and are not covered by warranty. Other replacement parts carry a 12-month warranty from the invoice date. Standard parts ship within 5 working days; diamond blades and spindle bearings may take up to 10 working days.

# PART 3 - TECHNICAL TROUBLESHOOTING

## Section 3.1 WP-400 error codes

Doc ref: TS-WP400-3.1 | Category: Technical Troubleshooting | Machine: WP-400

| Code  | Meaning                                      | Likely cause                                                | Action                                                                                                  |
|-------|----------------------------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
| E-101 | Air pressure low (below 5.0 bar)             | Compressor off, supply valve closed, regulator drift        | Check compressor and valve. Set regulator to 6.0 bar. Alarm clears above 5.5 bar.                       |
| E-102 | Dust extraction fault                        | Filter blocked (pressure drop above 1200 Pa) or fan stopped | Clean or replace cartridge WP4-FL-DE1. Check fan breaker.                                               |
| E-205 | Spindle overload                             | Dull blade, feed too high, slack belt                       | Reduce feed rate. Replace blade WP4-BL-300. Check belt tension (Section 2.1).                           |
| E-206 | Blade vibration / imbalance (above 4.5 mm/s) | Damaged blade, dirty flange, loose arbor nut                | Stop at once. Inspect blade and flanges. Replace blade per Section 4.1. Retighten nut to 60 Nm.         |
| E-310 | Y-axis following error                       | Dry rails, obstruction, encoder cable fault                 | Clear obstruction, grease rails (LG-2), check encoder cable. Recalibrate per Section 4.5 if it repeats. |
| E-311 | X-axis home switch not found                 | Switch dirty or misaligned                                  | Clean switch and check cable.                                                                           |
| E-410 | Emergency-stop circuit open                  | E-stop pressed or guard door open                           | Release E-stop, close door, press RESET.                                                                |

### Symptom guide - WP-400

| Symptom                      | Check                                               | Fix                                                       |
|------------------------------|-----------------------------------------------------|-----------------------------------------------------------|
| Edge chipping on melamine    | Blade condition, feed rate, scoring blade alignment | Replace blade, reduce feed 10-20%, realign scoring blade. |
| Cut sizes drift over the day | Y-axis calibration, belt, rail lubrication          | Recalibrate (Section 4.5) and grease rails.               |
| Machine will not start       | Air pressure, E-stop, extraction interlock          | Clear E-101, E-410 or E-102 first.                        |

## Section 3.2 SC-900 fault codes

Doc ref: TS-SC900-3.2 | Category: Technical Troubleshooting | Machine: SC-900

| Code | Meaning                        | Likely cause                                    | Action                                                                                           |
|------|--------------------------------|-------------------------------------------------|--------------------------------------------------------------------------------------------------|
| F-01 | Water flow low (below 8 L/min) | Clogged filter, valve partly closed, pump fault | Clean or replace SC9-WF-02. Open valve fully. Check pump.                                        |
| F-02 | Spindle stalled / overload     | Feed too fast, glazed blade                     | Reduce feed. Dress the blade with an abrasive block. Replace if segments are worn (Section 4.2). |
| F-03 | Bridge travel limit reached    | Slab positioned past limit                      | Reposition slab and re-home bridge.                                                              |
| F-07 | Slurry tray level high         | Drain blocked                                   | Flush tray and clear drain.                                                                      |

### Symptom guide - SC-900

| Symptom                    | Check                                               | Fix                                                                |
|----------------------------|-----------------------------------------------------|--------------------------------------------------------------------|
| Blade wobble or vibration  | Flange runout (limit 0.1 mm), bent blade, loose nut | Replace bent blade, clean flanges, torque nut to 80 Nm.            |
| Blade glazed, cuts slowly  | Wrong blade for the stone, feed too slow            | Dress with abrasive block; use correct blade (SC9-DB-350 granite). |
| Chipping on slab underside | Support, exit speed                                 | Support slab fully and slow down at exit.                          |

## Section 3.3 MC-250 fault codes

Doc ref: TS-MC250-3.3 | Category: Technical Troubleshooting | Machine: MC-250

| Code | Meaning                          | Likely cause                                            | Action                                                                           |
|------|----------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------|
| P-11 | Pilot arc failed / no arc        | Worn electrode or nozzle, low air pressure, poor ground | Replace consumable set (Section 4.3). Check 6.2 bar. Clean ground clamp contact. |
| P-14 | Air pressure low (below 5.5 bar) | Supply issue, blocked filter                            | Drain bowl, replace MC2-AF-10, check supply.                                     |
| P-22 | Torch height control fault       | Sensor not finding plate, damaged lead                  | Check lead, replace MC2-TH-HT if fault persists.                                 |

### Symptom guide - MC-250

| Symptom              | Check                                                | Fix                                                |
|----------------------|------------------------------------------------------|----------------------------------------------------|
| Dross on bottom edge | Cut speed, consumable wear                           | Adjust speed to table values; replace consumables. |
| Bevelled cut edge    | Torch squareness, nozzle wear                        | Square the torch; replace nozzle MC2-NZ-65.        |
| Arc goes out mid-cut | Ground clamp, rust or paint on plate, speed too high | Clamp to bare metal, reduce speed.                 |

## Section 3.4 GC-120 alarm codes

Doc ref: TS-GC120-3.4 | Category: Technical Troubleshooting | Machine: GC-120

| Code | Meaning                       | Likely cause                                               | Action                                                         |
|------|-------------------------------|------------------------------------------------------------|----------------------------------------------------------------|
| G-03 | Vacuum pressure low           | Worn seal strip, sheet not fully seated, pump filter dirty | Reseat sheet, clean pump filter, replace GC1-VB-02 if needed.  |
| G-05 | Scoring pressure out of range | Regulator drift                                            | Reset to the pressure for the glass thickness (Section 1.4.1). |
| G-08 | Cutting oil low               | Reservoir below MIN                                        | Refill with GC1-OL-1L.                                         |

### Symptom guide - GC-120

| Symptom                                         | Check                                            | Fix                                                         |
|-------------------------------------------------|--------------------------------------------------|-------------------------------------------------------------|
| Glass runs off the score line / breaks unevenly | Wheel wear, pressure too low, delay before break | Replace wheel, raise pressure per table, break within 30 s. |
| Shells along edge                               | Pressure too high, damaged wheel                 | Lower pressure; replace wheel (Section 4.4).                |
| No visible score line                           | Wheel not touching, oil empty                    | Check head height and refill oil.                           |

# PART 4 - RESOLUTION PROCEDURES

**SAFETY:** Before any maintenance or blade/consumable change, switch the machine off, lock out the main isolator per your site lock-out/tag-out procedure, and wait until all moving parts have stopped.

## Section 4.1 Replace the WP-400 saw blade

*Doc ref: RP-WP400-4.1 | Category: Technical Troubleshooting / Maintenance & Parts | Machine: WP-400*

1. Switch off and lock out the machine. Wait 2 minutes until the spindle has fully stopped.
2. Open the blade guard and insert the spindle lock pin.
3. Remove the arbor nut. The WP-400 arbor nut has a left-hand thread.
4. Remove the outer flange and old blade. Clean both flanges and the arbor.
5. Fit new blade WP4-BL-300 with the rotation arrow matching the spindle direction.
6. Refit the outer flange and tighten the arbor nut to 60 Nm.
7. Remove the lock pin, close the guard, remove lock-out, run at 1000 rpm for 30 seconds, then at full speed.
8. Reset the blade-hours counter. If E-206 returns, re-inspect the flanges for damage or debris.

## Section 4.2 Replace the SC-900 diamond blade

*Doc ref: RP-SC900-4.2 | Category: Technical Troubleshooting / Maintenance & Parts | Machine: SC-900*

1. Switch off, lock out, and close the water valve. Raise the bridge fully.
2. Lock the spindle and remove the flange bolt.
3. Remove the flange and blade. Check flange faces for burrs; runout limit is 0.1 mm.
4. Fit the new blade with the arrow in the direction of rotation. Tighten the flange bolt to 80 Nm.
5. Open the water, confirm 8 L/min, and run without cutting for 1 minute.
6. Make a first cut at half the normal feed to bed in the blade. Dress with an abrasive block if it glazes.

## Section 4.3 Replace MC-250 torch consumables

*Doc ref: RP-MC250-4.3 | Category: Technical Troubleshooting / Maintenance & Parts | Machine: MC-250*

1. Switch off the power source and disconnect it from the supply; shut off the air.
2. Unscrew the shield cap, then remove the nozzle, swirl ring and electrode in that order.
3. Replace the electrode if the pit is 1.5 mm deep. Always fit a matched set (electrode, nozzle, swirl ring, shield).
4. Lightly coat O-rings with silicone grease only. Do not use petroleum grease.
5. Refit parts hand-tight; do not use tools on the nozzle or shield.
6. Restore air at 6.2 bar and make a test cut on scrap.

## Section 4.4 Replace the GC-120 scoring wheel

*Doc ref: RP-GC120-4.4 | Category: Technical Troubleshooting / Maintenance & Parts | Machine: GC-120*

1. Switch off and lock out the machine. Move the scoring head to the service position.
2. Remove the retaining screw and take out the old wheel.
3. Fit a new wheel GC1-SW-6 and refit the screw finger-tight plus a quarter turn.
4. Switch on, check scoring pressure for the thickness, and make a test score on 4 mm scrap.
5. Break the test piece within 30 seconds. The edge should be clean without shells.

## Section 4.5 Recalibrate the WP-400 Y-axis

*Doc ref: RP-WP400-4.5 | Category: Technical Troubleshooting | Machine: WP-400*

1. Make sure rails are lubricated (grease LG-2) and the table is clear.

2. Home all axes, then open Diagnostics > Axis Calibration.
3. Make a 1000 mm test cut on scrap panel.
4. Measure the cut with a calibrated rule and enter the measured value.
5. Accept if the deviation is within 0.2 mm over 1000 mm. Otherwise repeat once; if it still fails, check the Y-axis belt and encoder cable.

# PART 5 - FREQUENTLY ASKED QUESTIONS

*Doc ref: FAQ-ALL-5 | Category: Mixed | Machine: All*

**[WP-400] Q: Why do I get E-101 right after restarting the compressor?**

A: The tank is still filling. E-101 clears when pressure rises above 5.5 bar; wait and press RESET.

**[WP-400] Q: How often should I change the saw blade?**

A: Typically every 80-120 operating hours, or sooner if edges chip. See the parts table in Section 2.3.

**[WP-400] Q: Can I run the saw with the extraction off for a quick cut?**

A: No. The interlock prevents blade start and bypassing it is not permitted.

**[SC-900] Q: Can I use dirty or recycled water?**

A: Only if it is filtered. Blocked water filters cause alarm F-01; clean SC9-WF-02 weekly.

**[SC-900] Q: How do I know the diamond blade is worn out?**

A: Segment height below 3 mm, or the blade cuts slowly even after dressing.

**[MC-250] Q: Why does my plasma arc go out halfway through a cut?**

A: Usually a poor ground connection, rust or paint on the plate, or speed set too high.

**[MC-250] Q: Which consumables should I keep in stock?**

A: At least two sets of electrode MC2-EL-65, nozzle MC2-NZ-65, swirl ring MC2-SW-65 and shield MC2-SH-65, plus a spare air filter MC2-AF-10.

**[GC-120] Q: Why does the glass break away from the score line?**

A: A worn wheel, low scoring pressure, or waiting more than 30 seconds before breaking.

**[GC-120] Q: Which oil should I use in the scoring head?**

A: Use GC1-OL-1L cutting oil only and keep the level above MIN.

**[All] Q: Are blades and nozzles covered by warranty?**

A: No. They are consumables. Other parts have a 12-month warranty (Section 2.4).